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Algebra Toolkit

The tools every algebra topic uses: the words, collecting like terms, and substituting numbers — especially negatives — without a slip.

THE TOPIC ON ONE MAP

Algebra Toolkit

01 — Terms and like terms

$$4y + 2x - 3 + 3x + 8$$

$$3a - a \times 4a + 2a$$

identity: find a, b

02 — Substitution

$$f = 3g + 7h, g = -5$$

$$s = ut + \frac{1}{2}at^2$$

“all the figures on your calculator”

03 — Substitute, then solve

$$A = 4bc: \text{find } c$$

$$h = 3t^2: \text{find } t$$

“if x doubles ...”

THE FIVE WORDS Know which one you are looking at:

ZONE 1

Term

$$4x - 3y^2$$

ZONE 2

Expression

$$4x - 3y \text{ (no } = \text{)}$$

ZONE 3

Equation

$$4x - 3 = 9 \text{ (solve it)}$$

ZONE 4

Formula

$$A = \pi r^2 \text{ (links letters)}$$

ZONE 5

Identity

$\equiv$ true for every x

01 Terms and Like Terms

A term is a number, letters, or both — and the sign in front belongs to it. Only **like terms** (same letters, same powers) can be added.

WHEN YOU SEE → YOU DO

“Simplify”

→ collect like terms: same letters, same powers

$-4a$ **after** $3a$

→ the sign moves with the term: $3a - 4a = -a$

$a \times 4a$

→ multiply first (BIDMAS): $4a^2$ — not a like term of a

≡ **“find a and b in the identity”**

→ expand, then match the x terms and the numbers

Circle each term with its sign

Group like terms

Add the numbers in front

WORKED MODEL • SIMPLIFY

Simplify $3a - 5b - a + 2b$.

1 $(3a - a) + (-5b + 2b)$

group like terms

ANSWER $2a - 3b$

WORKED MODEL • IDENTITY

$5(7x + 8) + 3(2x + b) \equiv ax + 13$. Find a and b .

1 $35x + 40 + 6x + 3b = 41x + 40 + 3b$

expand

2 x terms: $a = 41$; numbers: $40 + 3b = 13$

match

ANSWER $a = 41, b = -9$

× **TRAP** $\frac{6c + 10}{2}$ is **not** $3c + 10$. Divide **both** terms: $3c + 5$.

IDEA 01

Practise: Terms and Like Terms

My questions first, then selected practice.

Q1 DR ESLAM Collect

CH 2 • TOPIC 1 2 marks

Simplify $5x - 2y + 3x + 7y$

SELECTED PRACTICE

Q2 PRACTICE Practice • Easy

CH 2 • TOPIC 1 2 marks

3 Simplify $4y + 2x - 3 + 3x + 8$

(2 marks)

Q3**PRACTICE**

Practice · Easy

CH 2 · TOPIC 1

2 marks

5 Simplify $ab - 5g + 5ab - 2g$

(2 marks)**Q4****PRACTICE**

Practice · Easy

CH 2 · TOPIC 1

1 mark

7 Sofia is trying to simplify $\frac{6c + 10}{2}$

Her method is

divide $6c$ by 2

then add 10

Evaluate her method.

(1 mark)

Q5**PRACTICE**

Practice · Easy

CH 2 · TOPIC 1 2 marks

11 Simplify $5c - d - 3d - 2c$.

(2 marks)

Q6**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1 3 marks

3 a and b are positive values.

Show that $\frac{7a + 2b - 3a}{8a + 6b + 2a - b}$ always simplifies to the same value.

(3 marks)

Q7**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1**1 mark**

5 Amy has x beads.

Billy has three more beads than Amy.

Carly has four times as many beads as Billy.

Circle the expression for the number of beads that Carly has.

$4x + 3$	$3x + 4$	$4(x + 3)$	$x + 12$
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(1 mark)

02 Substitution

Replace every letter by its number **in brackets**, then use the calculator. Brackets protect the negatives.

WHEN YOU SEE → YOU DO

a negative value: $c = -5$

→ write (-5) — always in brackets

c^2 **with** $c = -5$

→ $(-5)^2 = 25$, not -25

$2qx^2$

→ square first, then multiply: $2 \times 3 \times (-5)^2$

“write down all the figures on your calculator”

→ no rounding at all

Write the formula

Numbers in brackets

Calculator

Answer with units

WORKED MODEL • NEGATIVES

$G = c^2 - 4c$. Find G when $c = -5$.

1 $G = (-5)^2 - 4(-5)$

brackets

2 $G = 25 + 20$

BIDMAS

ANSWER $G = 45$

× **TRAP** Typing -5^2 in the calculator gives -25 . Type $(-5)^2$.

IDEA 02

Practise: Substitution

My questions first, then selected practice.

Q8 DR ESLAM Negative squared

CH 2 • TOPIC 1 2 marks

Work out the value of $2x^2 - 4x$ when $x = -3$.

Q9 DR ESLAM Calculator trap

CH 2 • TOPIC 1 1 mark

Ali types -3^2 and gets -9 . What should he type to square -3 ?

SELECTED PRACTICE

Q10**PRACTICE**

Practice • Easy

CH 2 • TOPIC 1 2 marks

1



You can work out the amount of medicine, c *ml*, to give to a child by using the formula

$$c = \frac{ma}{150}$$

m is the age of the child, in months.

a is an adult dose, in *ml*.

A child is 30 months old.

An adult's dose is 40 *ml*.

Work out the amount of medicine you can give to the child.

(2 marks)

Q11**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 4 marks

- 1 (a)** The body mass index, B , for a person of mass m kg and height h metres is given by the formula

$$B = \frac{m}{h^2}$$

Usman has a mass of 50 kg.

He has a height of 1.57 m.

Work out Usman's body mass index.

Give your answer correct to one decimal place.

(2 marks)

- (b)** Tom's height is 1.80 m.
He wants his body mass index to be 21

Work out the mass that will give Tom a body mass index of 21

(2 marks)

Q12 **PRACTICE** Practice · Medium

CH 2 · TOPIC 1 2 marks

$$7 \quad w = 5y^2 - y^3$$

Work out the value of w when $y = -2$

$$w = \dots\dots\dots$$

(2 marks)

Q13 **PRACTICE** Practice · Medium

CH 2 · TOPIC 1 2 marks

$$11 \quad y = mx + c$$

Find the value of y when $m = -3$, $x = -2$ and $c = -8$.

$$y = \dots\dots\dots$$

(2 marks)

Q14 **PRACTICE** Practice · Medium

CH 2 · TOPIC 1 2 marks

$$13 \quad s = ut + \frac{1}{2}at^2$$

Find the value of s when $u = 5.2$, $t = 7$ and $a = 1.6$.

$s =$

(2 marks)

Q15 **PRACTICE** Practice · Hard

CH 2 · TOPIC 1 4 marks

$$7 \quad s = ut + \frac{1}{2}at^2$$

Find s when $t = 26.5$, $u = 104.3$, $a = -2.2$.

Give your answer in standard form, correct to 4 significant figures.

$s =$

(4 marks)

03 Substitute, Then Solve

Sometimes the letter you want is **not** the subject. Substitute what you know, then solve the equation that is left.

WHEN YOU SEE → YOU DO

" $A = 100, b = 2$, **work out c** " in $A = 4bc$ " → substitute, then solve: $100 = 8c$

$h = 3t^2$, "**find t when $h = 108$** " → $t^2 = 36$, $t = 6$ (or -6)

"**if x doubles, what happens to y ?**" → try numbers: $x = 1$ then $x = 2$

"**One solution is $x = 7$** " → put $x = 7$ in, find the constant

WORKED MODEL • FIND A LETTER INSIDE

$C = \frac{5(F - 32)}{9}$. Is 77°F below the 20°C minimum?

1 $C = \frac{5(77 - 32)}{9} = \frac{5 \times 45}{9}$

substitute

2 $C = 25$

calculator

ANSWER $25^\circ\text{C} > 20^\circ\text{C}$: not below the minimum

WHERE YOU WILL USE IT Every formula question in the course: rearranging (Topic 7), graphs, mensuration, kinematics.

IDEA 03

Practise: Substitute, Then Solve

My questions first, then selected practice.

Q16 **DR ESLAM** Then solve

CH 2 • TOPIC 1 2 marks

 $v = 5t - 3$. Find t when $v = 22$.**SELECTED PRACTICE****Q17** **PRACTICE** Practice • Easy

CH 2 • TOPIC 1 1 mark

$$8 \quad y = \frac{10}{x}$$

If the value of x doubles, what happens to the value of y ? Circle your answer.

$\div 2$	$\times 2$	$\div 5$	$\times 5$
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(1 mark)

Q18**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 2 marks

$$A = 4bc$$

2

$$A = 100$$

$$b = 2$$

Work out the value of c .**(2 marks)**

Q19**PRACTICE**

Practice • Medium

CH 2 • TOPIC 1 3 marks

3 You can change temperatures from °F to °C by using the formula

$$C = \frac{5(F - 32)}{9}$$

F is the temperature in °F.

C is the temperature in °C.

The minimum temperature in an elderly person's home should be 20 °C.

Mrs Smith is an elderly person.

The temperature in Mrs Smith's home is 77 °F.

Decide whether or not the temperature in Mrs Smith's home is lower than the minimum temperature should be.

(3 marks)

Q20**PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 3 marks

$$6 \quad h = 5t^2 + 2$$

(i) Work out the value of h when $t = -2$

[1]

(ii) Work out a value of t when $h = 47$

[2]

(3 marks)**Q21****PRACTICE**

Practice · Medium

CH 2 · TOPIC 1 1 mark

14 Complete the statement.

When $5x = 15$, $12x = \dots\dots\dots$

(1 mark)

Q22**PRACTICE**

Practice · Hard

CH 2 · TOPIC 1 3 marks

8 (a) One solution of the equation $ax^2 + a = 150$ is $x = 7$.

Find the value of a .

$a = \dots\dots\dots$
(2 marks)

(b) Find the other solution.

$x = \dots\dots\dots$ [1]
(1 mark)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------|--|
| 01 | Like terms | Same letters and powers; the sign moves with the term. Identity: match coefficients. |
| 02 | Substitution | Every number in brackets; $(-5)^2 = 25$; BIDMAS; no early rounding. |
| 03 | Then solve | Substitute what you know, solve for the letter left. |

BOTTOM LINE Brackets round every substituted number → BIDMAS → the sign belongs to the term after it.

Answers

Algebra Toolkit • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • TERMS AND LIKE TERMS

Q1	Dr Eslam • Collect	$8x + 5y$
Q2	Practice • Easy	$5x + 4y + 5$
Q3	Practice • Easy	$6ab - 7g$
Q4	Practice • Easy	wrong: 10 must be divided by 2 as well; $3c + 5$
Q5	Practice • Easy	$3c - 4d$
Q6	Practice • Hard	$\frac{4a + 2b}{10a + 5b} = \frac{2(2a + b)}{5(2a + b)} = \frac{2}{5}$
Q7	Practice • Hard	$4(x + 3)$

IDEA 02 • SUBSTITUTION

Q8	Dr Eslam • Negative squared	$2(-3)^2 - 4(-3) = 18 + 12 = 30$
Q9	Dr Eslam • Calculator trap	$(-3)^2 = 9$
Q10	Practice • Easy	$\frac{30 \times 40}{150} = 8 \text{ ml}$
Q11	Practice • Medium	(a) 20.3 (b) 68.04 kg
Q12	Practice • Medium	$w = 28$
Q13	Practice • Medium	$y = -2$
Q14	Practice • Medium	$s = 75.6$
Q15	Practice • Hard	1.991×10^3

IDEA 03 • SUBSTITUTE, THEN SOLVE**Q16** Dr Eslam • Then solve

$$22 = 5t - 3; t = 5$$

Q17 Practice • Easy

$$\div 2$$

Q18 Practice • Medium

$$100 = 8c; c = 12.5$$

Q19 Practice • Medium

$$C = 25^{\circ}\text{C} > 20^{\circ}\text{C}: \text{not lower}$$

Q20 Practice • Medium

$$(i) 22 (ii) t = 3 \text{ (or } -3)$$

Q21 Practice • Medium

$$12x = 36$$

Q22 Practice • Hard

$$(a) a = 3 (b) x = -7$$